

Age-related differences in the capacity and neuromuscular control of the foot core system during quiet standing

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Abstract

The foot core system is essential for upright stability. However, aging-induced changes in the foot core function remain poorly understood. The present study aimed to examine age-related differences in postural stability from the perspective of foot core capacity and neuromuscular control during quiet standing. Thirty-six older and 25 young adults completed foot core capacity tests including toe flexion strength, muscle ultrasonography, and plantar cutaneous sensitivity. The center of pressure (COP) and electromyography (EMG) of abductor hallucis (ABH), peroneus longus (PL), tibialis anterior (TA) and medial gastrocnemius (GM) were simultaneously recorded during double-leg and single-leg standing (SLS). EMG data were used to calculate muscle synergy and intermuscular coherence across three frequency bands. Compared to young adults, older adults exhibited thinner hallucis flexors, weaker toe strength, and lower plantar cutaneous sensitivity. The ABH thickness and plantar cutaneous sensitivity were negatively associated with the COP mean peak velocity in older adults, but not in young adults. Besides, older adults had higher cocontraction of muscles spanning the arch (ABH-PL) and ankle (TA-GM), and had lower beta- and gamma-band coherence of the ABH-PL and TA-PL during SLS. Foot core capacities became compromised with advancing age, and the balance control of older adults was susceptible to foot core than young adults in balance tasks. To compensate for the weakened foot core, older adults may adopt arch and ankle stiffening strategies via increasing muscle cocontraction. Furthermore, coherence analysis indicated that aging may increase the demand for cortical brain resources during SLS.

KEYWORDS

foot and ankle biomechanics, postural stability, intrinsic foot muscle, muscle synergy, intermuscular coherence

1 | INTRODUCTION

Aging-induced deterioration in postural stability increases the risk of falls and poses a serious threat to the health of older adults.¹ To maintain postural stability, a

well-stabilized foot is vital as it provides the base of support, concurrent somatosensory information,² and neuromuscular adjustments³ to counteract postural sway. The concept of the foot core system considers the arch as the core of the human foot and suggests that the active

subsystem (intrinsic and extrinsic foot muscles), passive subsystem (bone and ligaments), and neural subsystem (sensory receptors) interact with each other to stabilize the foot core.⁴

Adequate foot core capacities such as muscle strength and sensory inputs are required to prevent the foot from being mechanically unstable under varying loads.⁴ Impaired foot neural subsystem such as reduced plantar cutaneous sensitivity has been shown to lead to postural instability.⁵ And impaired foot active subsystem such as toe muscle weakness has been suggested as an independent predictor of falls in community-dwelling older adults.⁶ Several studies suggested that aging may induce alterations in foot core capacities such as muscle atrophy and cutaneous sensitivity decline.^{7,8} Improvements in the foot core capacities, such as strengthening intrinsic foot muscles, have shown beneficial effects on reducing fall incidence in the older population.⁹ However, the association between these foot core capacities and the balance performance of older adults remains poorly understood. Identifying this association would help formulate effective interventions to improve postural stability for older adults.

To compensate for the decline in foot core capacities, alterations in neuromuscular control strategies governing functional integration may occur to cope with postural sway. Neuromuscular features could be extracted from surface electromyographic (EMG) signals, such as muscle synergies,¹⁰ cocontraction,^{11,12} and intermuscular coherence,¹³ to uncover age-related neuromuscular control strategies during motor behaviors. According to muscle synergy theory, the central nervous system (CNS) controls specific muscle groups rather than a single muscle to perform motor tasks.¹⁰ Although intrinsic foot muscles are also activated and associated with the center of pressure (COP) sway during single-leg standing (SLS),^{3,14} they have yet to be included in muscle synergy analysis studies. To quantify the simultaneous activity of antagonist muscle pairs crossing the same joint, the cocontraction index (CI) is often calculated.^{12,15} Increased CI of ankle muscles (i.e., extrinsic foot muscles) was reported in older adults^{11,12} during balance tasks, as a mechanism of generating additional ankle stiffness. Whether the cocontraction of intrinsic-extrinsic foot muscle pairs would be altered with aging during balance tasks should be clarified.

Furthermore, intermuscular coherence between pairs of EMG signals has been used as a measure of shared cortical drive, suggesting the oscillatory synchronization of action potentials discharged by motor units,¹⁶ which is an attempt by the CNS to reduce the dimensionality and complexity of control.¹³ The human CNS drives muscle discharges at a number of

frequencies and these different activities may be considered as the characteristics of functional activities in distinct circuits.¹⁷ More specifically, the beta and gamma frequency band of intermuscular coherence has been suggested to originate from the motor cortex.^{17,18} Comparable to extrinsic foot muscles, intrinsic foot muscles are also innervated by the peroneal nerve, and tibial nerve. However, to the best of our knowledge, there is no published evidence of how healthy aging affects the neural drive of intrinsic-extrinsic foot muscle pairs during balance control. Establishing this effect would contribute to a more comprehensive understanding of balance neuromuscular control strategies in older adults.

Limited knowledge regarding how aging affects the role of the foot core system in balance control may impede the development of intervention programs to improve balance. Exploring how intrinsic and extrinsic foot muscles interact with each other in a challenging balance task may contribute to a more comprehensive understanding of neuromuscular control strategies in older adults. Thus, the purpose of this study was to determine age-related differences in postural stability from the perspective of the foot core capacity and neuromuscular control in quiet standing. It was hypothesized that older adults might have decreased foot core capacities and might adopt different neuromuscular control strategies during quiet standing.

2 | METHODS

2.1 | Participants

Thirty-six older adults without disabilities (mean $\pm$ SD: age = 68.8 ± 3.7 years, mass = 61.1 ± 8.8 kg, height = 162.4 ± 7.5 cm, BMI = 23.1 ± 2.5 ; 17 female) and 25 young adults (mean $\pm$ SD: age = 21.9 ± 1.9 years, mass = 56.7 ± 7.7 kg, height = 166.1 ± 8.8 cm, BMI = 20.5 ± 1.7 ; 14 female) were recruited in this study. The regular physical activity practice of older participants is shown in Table S1, with most of them having a moderate level of physical activity.¹⁹ None of the enrolled participants reported neurological disorders or falls in recent 5 years. All tests were conducted on the self-reported dominant leg and foot, identified by the response to the question “Which foot would you use to kick a ball with?”. This study was reviewed and approved by the Ethics Committee of the Seventh Affiliated Hospital of Sun Yat-sen University. All participants signed written informed consent for the experimental procedure, and the study was conducted in accordance with the Declaration of Helsinki.

2.2 | Assessment of foot core capacities

2.2.1 | Foot muscle morphology

Muscle thickness was measured using an ultrasound system (uSmart 3300 NexGen Ultrasound System, Terason, Burlington, USA) with a linear wideband array transducer (model: 15L4A Linear Array, Terason, Burlington, USA). The measured muscles were as follows: (1) extrinsic muscles: tibialis anterior (TA), peroneus longus (PL), flexor digitorum longus (FDL), flexor hallucis longus (FHL); (2) intrinsic foot muscles: digitorum brevis (FDB), flexor hallucis brevis (FHB) and abductor hallucis (ABH). Measurements were taken with the foot in the unloading position. A trained assessor obtained all images using a protocol by Crofts et al.²⁰ All ultrasound images were then processed in the ImageJ 2.3.0 software (National Institute for Health, Bethesda, USA) to measure muscle thickness. Our pilot study on five participants using the same protocol showed good intra-rater reliability, with intraclass correlation coefficients (ICC) ranging from 0.92 to 0.98.

2.2.2 | Foot muscle strength

The plantar flexion strength of the hallux and lesser toes was measured using a custom-made device, following the protocol described by Ridge et al.²¹ Before testing, participant practiced each test instrument to rule out potential learning effects. Thereafter, the participant performed the corresponding action to a maximal voluntary contraction against the dynamometer for 3 s, then relaxed as instructed by the rater. Each test was repeated three times.

2.2.3 | Plantar cutaneous sensitivity

Plantar cutaneous sensitivity thresholds at the first and fifth metatarsal head and medial longitudinal arch (MLA) were determined using Semmes Weinstein monofilaments (EXACTA Precision & Performance monofilaments, Stoelting, USA). Monofilaments were touched for 1.0–1.5 s at the test sites and 2.36–6.65 monofilaments were applied three times consecutively. When the participant felt the stimulus correctly in all three trials, the filament representing a specific force was noted as the participant's sensory threshold.

2.3 | Assessment of balance control

Two quiet standing postures (60-s double leg standing, DLS; and 30-s SLS) were performed three times with the

participant standing on a force platform (type: 9260AA6, Kistler, Winterthur, Switzerland, sampling at 1000 Hz) with their eyes open and focusing on a cross mark placed at eye level approximately 5 m in front of them and their arms folding across their chests. In the DLS task, participants were asked to place their feet shoulder-width apart. In the SLS task, once the raised foot was visually observed to touch the ground within 30 s by the researcher, this SLS trial was excluded. Sufficient resting time was provided between each trial. The COP signal during balance tasks was captured by the force plate. Surface EMG signals of the medial gastrocnemius (GM), TA, ABH and PL muscles were acquired with a surface EMG system (Trigno Wireless System, Delsys, Boston, USA, sampling at 2148 Hz) simultaneously. The locations of TA, PL and GM electrodes were determined according to the SENIAM project (<http://seniam.org/>). And, real-time ultrasonographic imaging was used to guide the surface electrode attachment to the ABH belly.²²

2.4 | Data analysis

All acquired data were analyzed using custom MATLAB routines (version R2021a, MathWorks, Natick, USA). The averaged COP, EMG, and strength values of three trials were taken for each participant who completed the three-times 30 s SLS task. The COP signal was low-pass filtered (fourth order, Butterworth, cut-off at 5 Hz) and was down-sampled to 100 Hz. Then its mean peak velocities (MPV) in the mediolateral (ML) and anteroposterior (AP) directions were calculated.²³ The raw EMG data were high-pass filtered (2nd order, Butterworth, cut-off at 30 Hz), and then rectified. To obtain an amplitude-normalized EMG envelope, the rectified EMG was smoothed with a low-pass (second order, Butterworth, cutoff at 12 Hz) filter,¹⁰ and normalized by their global maximum.

Muscle synergies were extracted from the matrix(M) consisting of amplitude-normalized EMG envelope using the Non-Negative Matrix Factorization (NNMF) algorithm.¹⁰ NNMF assumed that M is composed of a linear combination of a given number of muscle weights vectors (W) and the activation coefficient vectors. The optimal number of synergies was defined as the least number that provided 90% of the total variance accounted for (tVAF) between the measured and reconstructed matrix.¹⁰ Since balance control of SLS was not a typical cyclostationary process, the average recruitment level of the synergy (Recr) was computed as the average of the activation coefficient vector. In addition, the CI of agonist–antagonist foot muscle pairs was calculated using the amplitude-normalized EMG envelope.¹²

Intermuscular coherence was calculated from the rectified EMG by using the Welch method (window length

200 ms, overlap 100 ms) at the frequency of interest (4–60 Hz).¹³ Intermuscular coherence data was concatenated over six muscle pairs of all participants, and then NNMF was applied to divide it into three modules.¹³

2.5 | Statistical analysis

All statistical analyses were performed in IBM SPSS Statistics (version 26.0, IBM, Armonk, USA). The hypothesis of normality of the distribution was tested using the Shapiro–Wilk test. To determine the difference between older and young participants, independent sample *t*-test was performed for normally distributed data, and the Mann–Whitney *U*-test was performed if the normality hypothesis was rejected. Pearson's correlation coefficients were used to estimate the associations of postural stability with muscle measurements and with plantar cutaneous sensitivity. The hypothesis test was performed at a significance level of 0.05.

3 | RESULTS

3.1 | Foot core capacity

As shown in Table 1, older adults had significantly lower hallux and lesser toes flexion strength than young adults.

Muscle ultrasonography showed that older adults had significantly thinner hallux flexors (i.e., ABH, FHB, FHL) than young adults and that both groups had the comparable thickness of toe flexors (i.e., FDB, FDL). Furthermore, older adults had higher plantar cutaneous sensitivity thresholds at the 1st metatarsal head, 5th metatarsal head, and MLA than young adults.

3.2 | Postural stability

Figure 1A,B showed examples of COP displacements obtained from two representative participants from young and older adults, and similar patterns within the participants were visually acknowledged. Older adults had higher COP MPV in the ML and AP directions than young adults during DLS and SLS ($p < 0.01$, Figure 1C,D). Furthermore, thicker ABH muscle ($r = -0.501$, $p = 0.024$, Figure 1E) and FDB ($r = -0.482$, $p = 0.032$, Table S2) muscle were associated with lower COP AP MPV during SLS in older adults. Lower cutaneous sensitivity threshold of the MLA ($r = 0.610$, $p = 0.004$, Figure 1F), the 1st metatarsal head ($r = 0.455$, $p = 0.044$, Table S2) and fifth metatarsal head ($r = 0.495$, $p = 0.027$, Table S2), that is, higher cutaneous sensitivity, was associated with lower ML MPV during SLS in older adults. However, such associations were absent in young adults and other muscle parameters (Tables S2 and S3).

Variables	Young adults (<i>n</i> = 25)	Older adults (<i>n</i> = 36)	<i>p</i> -value
Foot strength (<i>N</i>)			
Hallux flexion strength	75.0 ± 21.2	47.1 ± 27.1	<0.001**
Toe flexion strength	50.5 ± 8.2	41.5 ± 15.4	0.005**
Muscle thickness (mm)			
ABH thickness	12.1 ± 1.6	10.9 ± 1.8	0.023*
FDB thickness	10.2 ± 1.6	9.4 ± 1.9	0.082
FHB thickness	15.9 ± 1.7	14.0 ± 2.3	0.001**
PL thickness	12.7 ± 2.2	12.3 ± 2.7	0.554
TA thickness	28.0 ± 3.0	26.4 ± 4.0	0.247
FDL thickness	10.7 ± 2.7	10.3 ± 2.6	0.625
FHL thickness	13.7 ± 3.4	11.7 ± 2.6	0.010*
Semmes–Weinstein monofilament light-touch thresholds			
The first metatarsal head	2.5 ± 0.31	4.0 ± 0.6	<0.001**
The fifth metatarsal head	2.5 ± 0.2	4.1 ± 0.5	<0.001**
MLA	2.4 ± 0.1	3.7 ± 0.5	<0.001**

Note: Results are indicated with mean ± standard deviation. The *p*-values (bold) were denoted by

* $p < 0.05$, ** $p < 0.01$.

Abbreviations: ABH, abductor hallucis; FDB, digitorum brevis; FDL, flexor digitorum longus; FHB, flexor hallucis brevis; FHL, flexor hallucis longus; MLA, medial longitudinal arch; PL, peroneus longus; TA, tibialis anterior.

TABLE 1 Foot muscle strength, muscle thickness, and plantar cutaneous sensitivity thresholds for young and older adults.

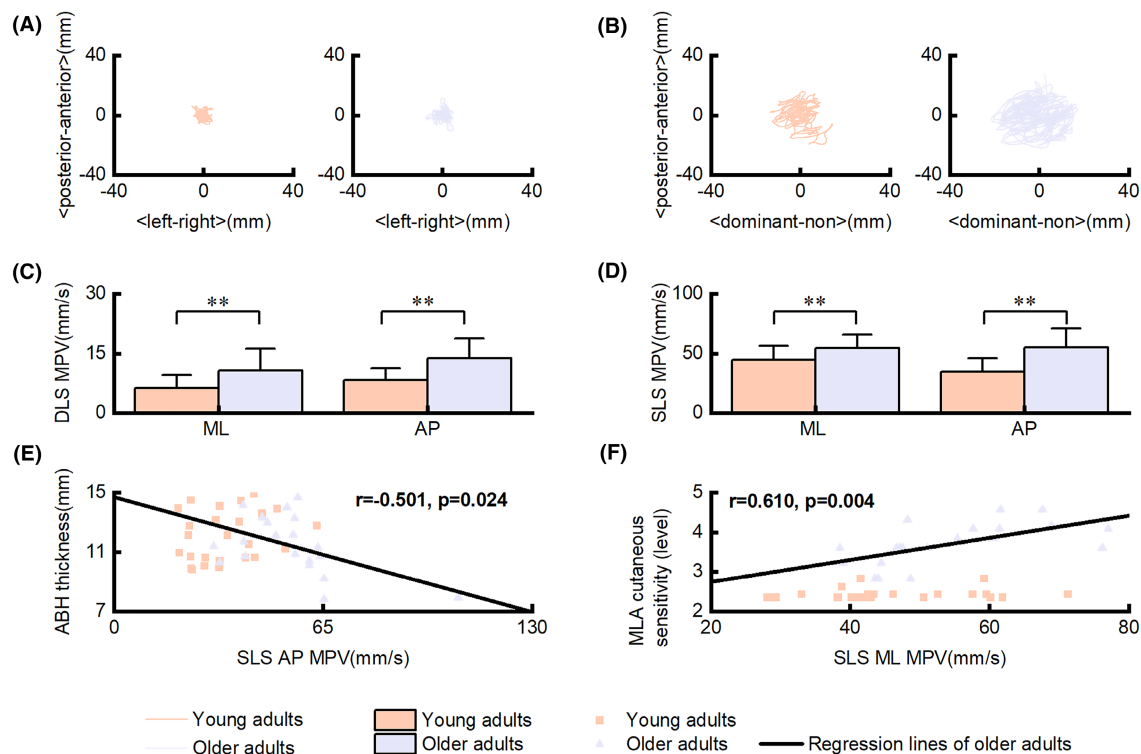


FIGURE 1 Representative examples of COP displacements during DLS (A) and SLS (B). COP MPV during the DLS (C) and SLS (D). The associations between AP MPV and ABH thickness (E), and between ML MPV and MLA sensitivity during SLS of older adults (F). ABH, abductor hallucis; AP MPV, mean peak velocity in anteroposterior direction; COP, center of pressure; DLS, double-leg standing; ML MPV, mean peak velocity in mediolateral direction; MLA, medial longitudinal arch; SLS, single-leg standing. The p values were denoted by * $p < 0.05$, ** $p < 0.01$.

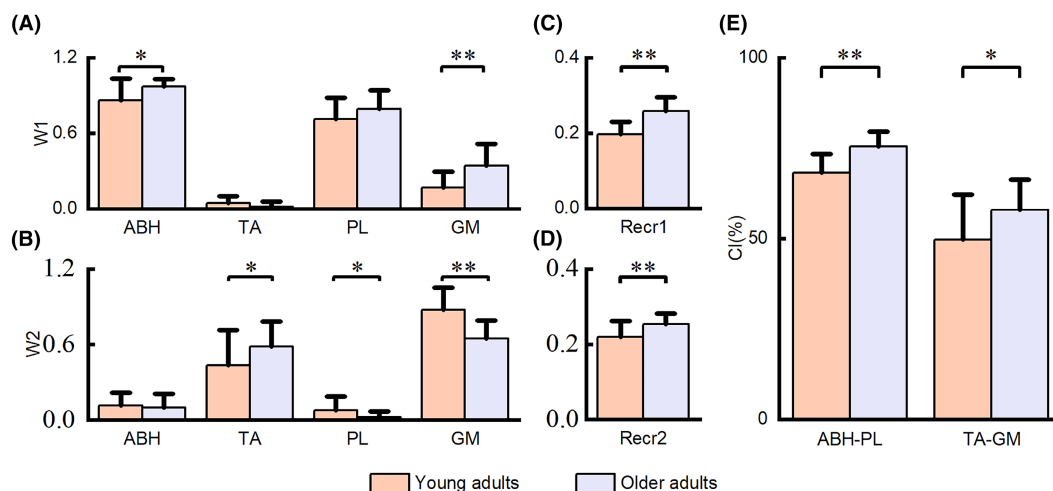


FIGURE 2 The weight vector of the first (A) and second (B) muscle synergies. The average recruitment level of the first (C) and second (D) muscle synergies. Muscle cocontraction index of ABH-PL and TA-GM (E). ABH, abductor hallucis; CI, cocontraction index; GM, medial gastrocnemius; PL, peroneus longus; SLS, single-leg standing; TA, tibialis anterior. The p values were denoted by * $p < 0.05$, ** $p < 0.01$.

3.3 | Neuromuscular control features

Two muscle synergies were necessary to reconstruct the EMG data with a tVAF value of 93.3% in older adults who completed the SLS task, and 92.6% in young adults, respectively (Figure 2A–D). The first muscle synergy was

dominated by the ABH and PL, and the second muscle synergy was dominated by TA and GM. In the first muscle synergy, the ABH and GM had a higher contribution in older adults than in young adults (Figure 2A). In the second muscle synergy, the TA had a higher contribution, while the GM and PL had a lower contribution in older

adults (Figure 2B). Older adults showed higher average recruitment levels of two muscle synergy modules during SLS than young adults (Figure 2C,D). The CI result showed that older adults had higher cocontraction of ABH-PL and TA-GM muscle pairs during SLS than young adults ($p=0.015$ and $p<0.001$, respectively, Figure 2E).

The intermuscular coherence was divided into 4–15 (alpha band), 15–30 (beta band), and 30–60 Hz (gamma band) (see Figure 3A). Older adults showed higher alpha-band coherence of almost all muscle pairs than young adults (see Figure 3B,C). Older adults also showed lower beta-band coherence of ABH-PL, TA-GM and TA-PL and lower gamma-band coherence of ABH-PL and TA-PL than young adults (see Figure 3B,C). The intermuscular coherence results of the DLS were shown in Figure S1, which were comparable to the results of the SLS.

4 | DISCUSSION

In the present study, we examined the effects of aging on foot core capacities and its neuromuscular control features during static balance tasks. Our results showed an aging-induced decline in the muscular and sensory capacities of the foot core, which may explain the worse balance during DLS and SLS in older adults. Correlation analysis results suggested that the balance control of older adults may be susceptible to several variables that related to the foot core capacity including the plantar cutaneous sensitivity and the ABH morphology than young adults.

Furthermore, our results confirmed the hypothesis that aging affected neuromuscular control features when coping with postural sway during SLS. Compared to young adults, older adults had a different spatial composition of muscle synergies, an increase in cocontraction of muscles spanning the arch and ankle joint, and a decrease in the beta-band coherence of ABH-PL, TA-GM, and TA-PL during SLS. These results indicated that older adults may adopt a less efficient neuromuscular control strategy, such as higher muscular costs and higher cortical resources to compensate for the weakened foot core system.

4.1 | Aging effects on the foot core capacity

4.1.1 | Muscle and sensory capacities

Due to the specific morphological and anatomical properties of distal foot muscles, they were more sensitive to body sway than proximal muscles.²⁴ Strong correlations were observed between the size of foot muscles and plantar flexion strength of hallux and toes.⁷ Older adults showed 37% and 18% lower plantar flexion strength of the hallux and lesser toes, respectively than young adults in the current study, which was consistent with a previous study.⁷ The decline of hallux flexion strength was further supported by the age-related differences in the thickness of hallux flexors that attach intrinsically and extrinsically to the foot (10% thinner ABH, 8% thinner FHB and 15%

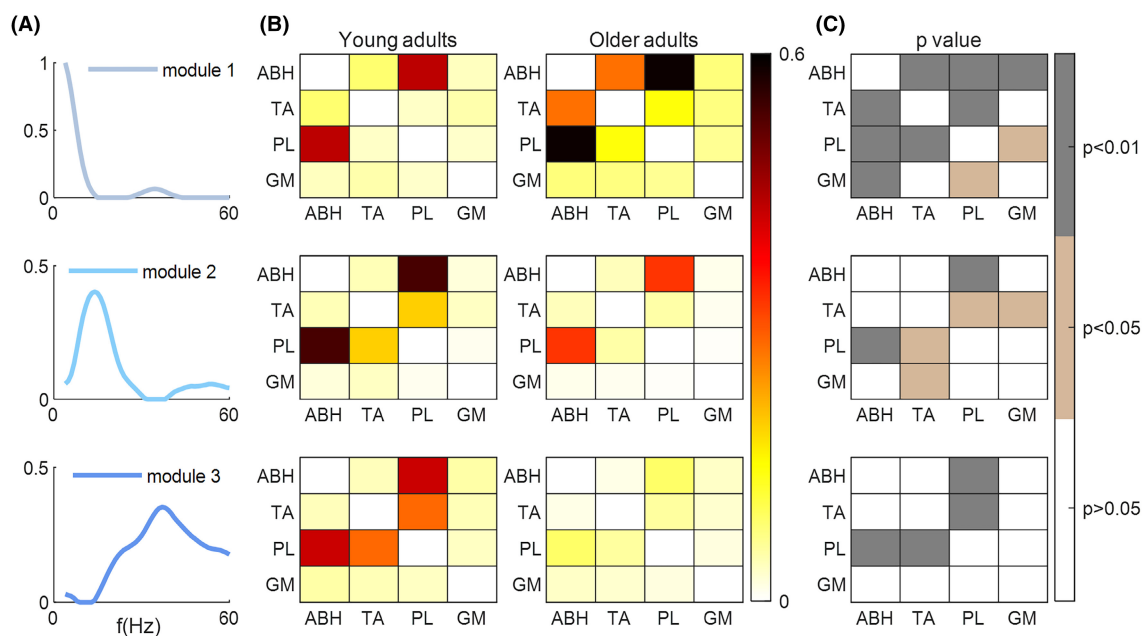


FIGURE 3 Three frequency components 4–15, 15–30, and 30–60 Hz (A) and coherence weights (B) of intermuscular coherence during SLS. The p value for comparison of intermuscular coherence between young and older adults (C). ABH, abductor hallucis; GM, medial gastrocnemius; PL, peroneus longus; SLS, single-leg standing; TA, tibialis anterior.

thinner FHL of older vs. young adults), which agreed with a 12-year longitudinal study on aging.²⁵

Furthermore, our results showed a decline in cutaneous sensitivity of the first and fifth metatarsal heads and the MLA due to aging, which was in line with previous studies.⁸ Low-threshold mechanoreceptors distributed at these locations had high sensitivity and extremely short latency, which allowed the fast sensation of subtle body sway.² The reduced sensitivity would limit the sensory input transferred to the CNS to formulate timely motor commands to counteract excessive body sway.^{4,8} Consequently, the decline in muscular and sensory capacities of the foot core may be a potential factor contributing to poor balance performance in older adults.

4.1.2 | Association between foot core capacities and balance

The COP MPV in the ML and AP directions serve as sensitive indicators of balance performance.²³ As expected, the COP MPV could discriminate older from young adults in both DLS and SLS. Considering that the ABH is the largest MLA-supporting intrinsic foot muscle, and that the arch region is the most sensitive plantar region,⁵ the current study further analyzed their association with the COP MPV during SLS. In older adults, better AP steadiness (lower AP MPV) was associated with thicker ABH, and better ML steadiness (lower MP MPV) was associated with a lower cutaneous sensitivity threshold of the MLA. The latter result was consistent with a recent study conducted on a larger sample (206 older adults).⁵ Interestingly, such correlation was not observed in young adults. Literature has documented age-related differences in visual contribution, vestibular perceptual thresholds, and ankle strength to postural stabilization.^{26–28} Here, our results indicated that aging might also affect the contribution of foot core capacities to balance maintenance, with balance control of older adults being susceptible to the foot core system. As such, strengthening the foot core might cause more pronounced balance improvements for older than young adults.

4.2 | Aging effects on the neuromuscular control

To compensate for the decline in foot core capacities above, alterations in neuromuscular control may occur during balance tasks. Compared to DLS, SLS is a challenging balance task, which is highly integrative and demanding.^{3,29} In the current study, only 56% of older adults accomplished 30s SLS, which agreed with a previous study on 1827 older adults.³⁰ The reduced ability to stand on one

leg has been suggested as a particularly strong predictor of injurious falls.³¹ Therefore, determining the neuromuscular characteristics of older adults who had preserved balance during SLS may yield insights into age-related neuromuscular control strategies.

4.2.1 | Muscle synergies and cocontraction

Each muscle synergy has a fixed spatial composition (motor module) and time-varying coefficients, and the spatial composition is quite robust across different motor tasks.³² The number of motor modules to perform a motor task could reflect neuromuscular complexity, and fewer motor modules were reported in older adult fallers and motor-impaired individuals.³³ In the current study, two muscle synergies were sufficient to reconstruct the measured EMG activity during SLS, which was consistent with a previous study that recorded far more muscles (13 muscles).³⁴ The spatial composition of synergy 1 was dominated by ABH and PL, which function to elevate/lower the arch. The composition of synergy 2 was dominated by TA and GM, which function to dorsiflex/plantarflex the ankle joint. Based on the anatomical features of those muscles,^{4,35} synergy 1 could be interpreted as an arch stabilizer and synergy 2 as an ankle stabilizer in this study. Our results showed that healthy aging would not alter the number of muscle synergies to preserve static and walking balance, which agreed with previous studies.^{33,36} However, age-related differences in the spatial composition were observed in this study, with a higher contribution of the ABH in the arch stabilizer synergy in older than young adults. This neuromuscular control strategy further confirmed the important role of the ABH in balance control in older adults.

As these two synergies were dominated by two pairs of opposing muscles, their CI was calculated as well for better comparison with other studies. We found an age-related increase in the cocontraction of ankle muscles, which was similar to previous studies.^{11,26} As the EMG of both intrinsic and extrinsic foot muscles were recorded, we further found an age-related increase in the cocontraction of arch muscles (ABH-PL). This increase might result from limited strength capacity, and has been associated with decreased spinal reciprocal inhibition and altered cortical function in older adults.³⁷ However, it should be noted that muscle cocontraction is at the expense of elevated energetic cost and body rigidity.¹⁵ Fortunately, a randomized controlled trial showed that an 8-week balance training program could not only exert beneficial effects on balance performance but also lower cocontraction of TA-Soleus muscles during SLS in older adults.¹² Further study is required to examine whether a long-term foot core strengthening

intervention would reduce the cocontraction of intrinsic-extrinsic foot muscles.

This increased cocontraction may also reflect a biomechanical joint stiffening strategy.²⁹ Increased cocontraction of agonist-antagonist muscles across the ankle has been suggested as an ankle-stiffening strategy,^{11,26} which allowed quick reactions to postural destabilization and froze the joint to counteract increases of body sway in challenging postural tasks.¹¹ Similarly, the high cocontraction of the ABH-PL pair in older adults may indicate an arch-stiffening strategy by freezing their arch. Apart from muscle cocontraction, strengthening intrinsic foot muscles has also been shown to increase arch stiffness³⁸ and balance performance.⁹ Understanding how arch-stiffening contributed to balance may help guide a proper balance intervention for older adults. Further studies are needed to determine the association between arch stiffness and balance performance and to formulate an energetic cost-effective way to stiffen the arch.

4.2.2 | Intermuscular coherence

Our results showed that older adults not only had higher activation levels during the SLS, but also had higher alpha band intermuscular coherence, which agreed with a previous study showing that the intermuscular coherence in the low-frequency band was positively related to muscle activities.³⁹ While intermuscular coherence at higher frequency bands (beta and gamma) is considered to have a cortical origin.¹⁸ Thus, the lower beta- and gamma-band coherence found in older adults suggested an aging-induced reduction in common neural drive to muscles to maintain SLS balance, which was consistent with previous findings on cortico-muscular coherence.⁴⁰ This reduction may be associated with age-related functional changes in the nervous system, such as an aging-induced decline in presynaptic inputs to the motor neuron pool.¹⁶ Furthermore, lower beta-band coherence of muscles also indicated an increased demand for cortical brain resources to maintain stability,⁴¹ indicating that older adults may adopt a less efficient way to cope with postural sway than young adults. It is worth knowing that this beta-band coherence could be adapted via muscle training. For instance, the lifting toes game has been shown to increase the beta-band coherence of extrinsic foot muscle pairs during gait in older adults.⁴² Whether intrinsic foot muscle training could increase the coherence of intrinsic-extrinsic foot muscle pairs in older adults should be determined in the future.

A recent study reported that the beta-band coherence of synergistic muscle pairs of ankle and knee joints (such as TA-PL) reduced with aging, while the antagonist muscle pairs (such as TA-gastrocnemius lateralis) remained unchanged during walking.¹⁶ The former result was consistent

with our results on ankle muscles during SLS. This may be a neuromuscular sign of age-induced ankle dorsiflexion weakness, an important physiological marker predicting mobility loss reported by a nine-year follow-up study.⁴³ In contrast with the latter result, the beta-band coherence of antagonist muscle pairs (TA-GM) was also reduced with aging. This might be due to differences in the motor task and muscle pairs. In addition, our result showed that the beta-band coherence of the antagonist muscle pair across the arch (i.e., ABH-PL) was significantly reduced with aging. This indicated that independent neural commands might be delivered to the ABH and PL to control the arch. These independent neural commands may weaken the confrontation between antagonistic muscles. Moreover, increasing studies analyzed the muscle network based on intermuscular coherence to determine the neural circuitry driving the musculoskeletal system during postural control.¹³ Our results suggested that the contribution of intrinsic foot muscles should be considered in future muscle network studies.

4.3 | Limitations

Several limitations of our work should be noted. First, the hypothesis of arch-stiffening should be considered carefully due to the lack of kinematic data and EMG of deeper muscles to quantify the real arch stiffness in this study. Future studies should consider recording the activity of deeper muscles, such as the tibialis posterior, to examine age-induced alterations in foot, and ankle function. Second, all participants of the study were healthy residents who did not report falls in the recent 5 years, which may limit the generalizability of the results to populations with a higher risk of falls. Third, the hallux flexion strength was related to FHL, FHB, ABH, and adductor hallucis, but the morphology of adductor hallucis was not measured in this study. In addition, only the EMG of GM was collected to represent the activity of ankle plantarflexors, future study should also examine soleus and gastrocnemius lateralis.

In summary, our findings suggested a decline in the foot core capacities and an altered neuromuscular control strategy during SLS with aging. To compensate for the weakened foot core capacities, older adults may adopt arch and ankle joint stiffening strategies via increasing muscle co-contraction (ABH-PL, TA-GM). Importantly, the common neural drive of muscles across the arch and ankle decreased during SLS, which may lead to the increasing demand for cortical brain resources. Besides, sensitivity of plantar cutaneous and muscle morphology were negatively associated with COP sway during SLS in older adults but not in young adults. It indicated that the balance control of older adults was susceptible to these foot core capacities for balance. Findings of this study may have implications for

understanding the impact on aging-induced deterioration in foot core system during balance control.

PERSPECTIVE

This study focused on aging-related differences in the capacity and control of foot core during standing. Recent studies indicated that foot core system play an important role in balance control.^{5,9,14} Our study found a decline in the foot core capacities and an altered neuromuscular control strategy during SLS with aging. These results indicated that older adults may adopt a less efficient neuromuscular control strategy, such as higher muscular costs and higher cortical resources to compensate for the weakened foot core system. This work also confirmed that the balance control of older adults was susceptible to the foot core system for balance. Preventive interventions that enhance the capacity and control of the foot core system should be encouraged in older adults.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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